Numerics

The ODE integrations for token sample paths (eg. Figure 4) and Watanabe–Strogatz parameters are performed using scipy's native 'solve_ivp' solver. We use the default Runge-Kutta method of order 4 (5), where error is controlled by the fourth order but steps are performed at the fifth order. Since rescaling β is equivalent to rescaling time, we always take $\beta = 1$ unless otherwise specified.

The Ott–Antonsen phase portraits are drawn by computing a 100×100 grid of sample velocities equally spaced in the domain $(\rho, \phi) \in [0.05, 1] \times [-\pi, \pi]$. In general the dynamics (3.1) are not defined at $\rho = 0$ because of the division by ρ in the $\dot{\phi}$ term; sampling from 0.05 avoids this issue. The phase portraits are drawn with matplotlib's streamplot function. The streamlines only show the direction of flow, not magnitude.

In the experiments section, tokens $\{x_k(0)\}_{k=1}^n$ are initialized uniformly on $\mathbb{S}^{d-1}$ via normalized Gaussian draws, and the ODE is integrated using the RK45 method with tolerances $\texttt{atol} = 10^{-8}$, $\texttt{rtol} = 10^{-6}$.

Generalization to higher dimensions. In the 2D analysis, the matrices A and V are specified entry-by-entry (each is 2×2). For $d > 2$, specifying d^2 entries individually is impractical, and more importantly, the case conditions are stated in terms of spectral properties of A and V (eigenvalues, definiteness of symmetric parts) rather than individual entries. We therefore construct the matrices to satisfy the relevant spectral condition in each case, using a random orthogonal basis to ensure that no special coordinate direction is preferred.

Throughout, we write $Q_{\mathrm{orth}} \in \mathbb{R}^{d \times d}$ for a Haar-random orthogonal matrix (sampled via the QR decomposition of a standard Gaussian matrix), $\mathrm{diag}(\lambda)$ for the diagonal matrix with entries $\lambda \in \mathbb{R}^d$, and $\mathrm{Skew}(d)$ for the space of $d \times d$ skew-symmetric matrices.

Case 1: $V = I$, A Arbitrary The relevant spectral conditions on A are stated in terms of $\mathrm{tr}(A)$, $\det(A)$, and the definiteness of the symmetric part $\frac{A+A^\top}{2}$. We construct A for each regime as follows.

Regime 1: Full clustering via $\mathrm{tr}(A) > 0$. We build A with a symmetric positive definite base plus a small skew-symmetric perturbation that preserves the trace condition while breaking symmetry:

$$A = Q_{\mathrm{orth}}\, \mathrm{diag}(\lambda)\, Q_{\mathrm{orth}}^\top + \varepsilon S, \qquad \lambda_i \sim \mathrm{Uniform}(0.5, 2.0), \quad S \in \mathrm{Skew}(d), \quad \varepsilon = 0.2\, \sigma_{\mathrm{off}}, \tag{1}$$

where $\sigma_{\mathrm{off}} \geq 0$ is a user-controlled off-diagonal scale parameter and S is sampled by antisymmetrizing a standard Gaussian matrix. Since $\lambda_i > 0$ for all i, we have $\mathrm{tr}(A) = \sum_i \lambda_i > 0$, and the skew perturbation does not change the trace (as $\mathrm{tr}(S) = 0$ for any skew-symmetric S). The resulting A is a non-symmetric matrix with positive trace and a positive definite symmetric part.

Regime 2: Full clustering via $\mathrm{tr}(A) \leq 0$ and $\det(A) \leq 0$. We construct A with one small positive eigenvalue $\lambda_+ > 0$ and $d-1$ negative eigenvalues $\lambda_2, \ldots, \lambda_d < 0$, chosen so that $\mathrm{tr}(A) = \lambda_+ + \sum_{i \geq 2} \lambda_i \leq 0$. Since the product $\det(A) = \lambda_+ \prod_{i \geq 2} \lambda_i$ is the product of one positive number and $d-1$ negative numbers, its sign is $(-1)^{d-1}$: negative for even d (satisfying $\det(A) \leq 0$ directly) and positive for odd d (violating the condition for A itself, but satisfied for $A^\top$ or via a sign adjustment). In practice we set

$$\lambda_+ \sim \mathrm{Uniform}(0.3, 0.8), \qquad \lambda_i \sim -\mathrm{Uniform}\left(\frac{\lambda_+}{d-0.5}, \frac{1.5\, \lambda_+}{d-0.5}\right) \text{ for } i \geq 2, \tag{2}$$

so that $|\sum_{i \geq 2} \lambda_i| > \lambda_+$ and the trace is non-positive. The skew perturbation is added as in Regime 1, and we verify post-hoc that the resulting A satisfies $\mathrm{tr}(A) \leq 0$, $\det(A) \leq 0$, and has at least one eigenvalue with non-negative real part.

Regime 3: Partial synchronization via $(A+A^\top)/2 \prec 0$. We build A with a negative definite symmetric part:

$$A = Q_{\mathrm{orth}}\, \mathrm{diag}(-\lambda)\, Q_{\mathrm{orth}}^\top + \varepsilon S, \qquad \lambda_i \sim \mathrm{Uniform}(0.5, 2.0), \tag{3}$$

so that the symmetric part $\frac{A+A^\top}{2} = Q_{\mathrm{orth}}\, \mathrm{diag}(-\lambda)\, Q_{\mathrm{orth}}^\top$ is negative definite. The skew perturbation εS does not affect the symmetric part (since $S + S^\top = 0$), so $(A + A^\top)/2 \prec 0$ is guaranteed regardless of ε.

In this regime, the OA analysis predicts convergence to a partial synchronization state $\rho_{eq} \in (0,1)$. For $d > 2$, the explicit formula (C.13) for ρ_{eq} in terms of the entries of A is not available; instead we observe the plateau value of $\hat{\mathcal{R}}_2$ numerically, which provides empirical evidence that an analogous partial synchronization equilibrium exists in higher dimensions.

Case 2: $A = I$, V Symmetric We set $A = I_d$ and construct V as a symmetric matrix with a prescribed dominant eigenvalue $\lambda_1 \in \mathbb{R}$ and $d-1$ subsidiary eigenvalues drawn randomly:

$$V = Q_{\text{orth}}\,\text{diag}(\lambda)\,Q_{\text{orth}}^\top, \qquad \lambda_1 = \lambda_{\max}\ \text{(user-specified)}, \qquad \lambda_i \sim \mathcal{N}(0, \sigma_{\text{other}}^2)\ \text{for}\ i \geq 2, \tag{4}$$

where the subsidiary eigenvalues are clamped to ensure λ_1 is strictly the largest eigenvalue. Specifically:

- If $\lambda_{\max} > 0$: eigenvalues λ_i for $i \geq 2$ are clamped to $(-0.9\lambda_{\max}, 0.9\lambda_{\max})$, so $\lambda_{\max}$ is the unique dominant eigenvalue.
- If $\lambda_{\max} = 0$: all subsidiary eigenvalues are set to $-|\lambda_i|$, so all eigenvalues are non-positive.
- If $\lambda_{\max} < 0$: subsidiary eigenvalues are clamped to $(-\infty, \lambda_{\max} - \varepsilon)$, so all eigenvalues are strictly negative and $\lambda_{\max}$ is the least negative.

The orthogonal matrix Q_{orth} determines the eigenvectors of V; since it is Haar-random, the dominant eigenvector v_1 (the preferred clustering direction) is uniformly distributed on $\mathbb{S}^{d-1}$, with no alignment to any coordinate axis.

Case 3: $A = I$, V Rotational In 2D, the rotational case uses $V = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$, which can be written as $V = aI_2 + bJ_2$ where $J_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ is the standard 2×2 rotation generator. The natural generalization to dimension d replaces J_2 with a block-diagonal skew-symmetric matrix J_d composed of $\lfloor d/2 \rfloor$ identical 2×2 blocks:

$$V = aI_d + bJ_d, \qquad J_d = \text{blkdiag}\underbrace{\left(\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \ldots, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\right)}_{\lfloor d/2 \rfloor \text{ blocks}}, \tag{5}$$

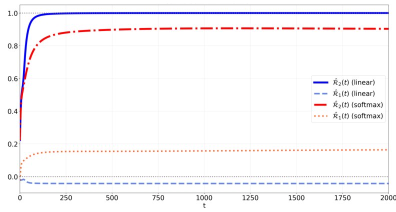
with $J_d[d,d] = 0$ when d is odd (zero-padding the last diagonal entry). This construction preserves the two key properties of the 2D case: (i) the symmetric part of V is $aI_d \succ 0$ for $a > 0$, which drives clustering; and (ii) the skew-symmetric part bJ_d induces a rotation in each of the $\lfloor d/2 \rfloor$ coordinate planes, controlling the phase velocity of the cluster without affecting whether clustering occurs. The 2D OA prediction $\dot{\rho} = \frac{a\beta}{2}(1-\rho^2)\rho$, $\rho \to 1$ for $a > 0$, is expected to generalize to $d > 2$ since the clustering force is governed by the symmetric part aI_d of V, which is isotropic and independent of d.

Case 4: $A = I$, V Hamiltonian In 2D, the Hamiltonian case uses $V = \begin{pmatrix} a & b \\ -b & -a \end{pmatrix}$, whose eigenvalues are $\pm\sqrt{a^2 - b^2}$ (real for $a > b$, purely imaginary for $a < b$, zero for $a = b$). The bifurcation at $a = b$ separates the clustering regime from the oscillatory regime. For $d > 2$, we generalize by taking V to be block-diagonal with $\lfloor d/2 \rfloor$ identical 2×2 Hamiltonian blocks:

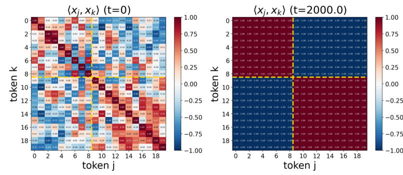
$$V = \text{blkdiag}\underbrace{\left(\begin{pmatrix} a & b \\ -b & -a \end{pmatrix}, \ldots, \begin{pmatrix} a & b \\ -b & -a \end{pmatrix}\right)}_{\lfloor d/2 \rfloor \text{ blocks}}, \tag{6}$$

with $V[d,d] = 0$ when d is odd. Each 2×2 block independently exhibits the same bifurcation structure as in 2D, so the d-dimensional system inherits the three qualitative regimes: complete clustering ($a > b$), oscillatory dynamics ($a < b$), and marginal behavior ($a = b$). This construction is natural in the sense that it reduces exactly to the 2D case when $d = 2$, and the bifurcation condition $a \lesseqgtr b$ is dimension-independent. However, for $d > 2$ with d odd, the unpaired last dimension receives no Hamiltonian drive ($V[d,d] = 0$), which may slightly modify the dynamics in that coordinate; we expect this effect to be negligible for large d.

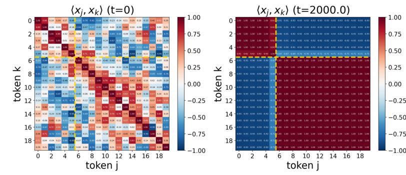
Case 4 More Experiments for $v_{11} = v_{12}$ Case: For different token initializations.



(a) Evolution of the clustering diagnostics $\hat{\mathcal{R}}_1(t)$ and $\hat{\mathcal{R}}_2(t)$ for the LSA (blue) and USA (red) models.

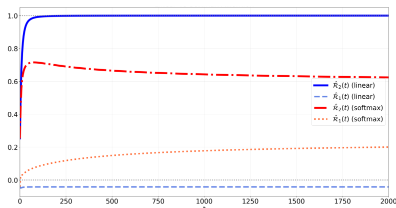


(b) Gram matrices for the LSA model.

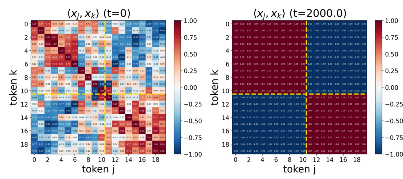


(c) Gram matrices for the USA model. (Clustering almost at antipodal points.)

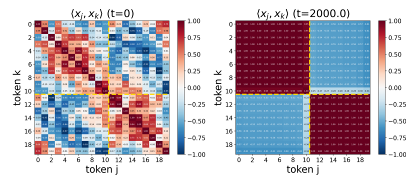
Figure 1



(a) Evolution of the clustering diagnostics $\hat{\mathcal{R}}_1(t)$ and $\hat{\mathcal{R}}_2(t)$ for the LSA (blue) and USA (red) models.

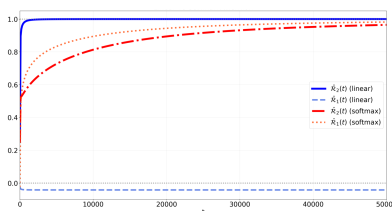


(b) Gram matrices for the LSA model.

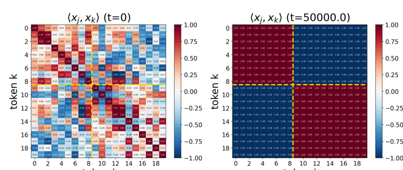


(c) Gram matrices for the USA model. (Clustering at two points but not antipodal points.)

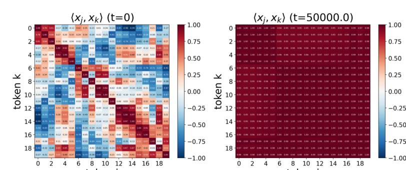
Figure 2



(a) Evolution of the clustering diagnostics $\hat{\mathcal{R}}_1(t)$ and $\hat{\mathcal{R}}_2(t)$ for the LSA (blue) and USA (red) models.



(b) Gram matrices for the LSA model.



(c) Gram matrices for the USA model. (Collapse into single point.)

Figure 3